

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	AIML-C2
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: **MULTI-AGENT AI RESEARCH PAPER ASSISTANT**

ABSTRACT

The rapid growth of research publications has made it increasingly difficult for students, researchers, and academicians to efficiently analyze and understand scientific literature. Reading multiple research papers, extracting key information, preparing presentations, identifying references, and understanding implementation details are time-consuming tasks. This project proposes a **Multi-Agent AI Research Paper Assistant** that leverages **Large Language Models (LLMs)** and **Retrieval-Augmented Generation (RAG)** to automate research paper analysis and knowledge generation.

The proposed system accepts one or more research papers in PDF format as input. A **Manager Agent** analyzes the user's request and coordinates multiple specialized AI agents to perform different tasks collaboratively. The **Summary Agent** generates concise summaries of various sections of the paper, the **Reference Agent** extracts citations, authors, keywords, and bibliography information, the **Presentation Agent** automatically creates presentation slides based on the paper, the **Email Agent** drafts professional emails for sharing research findings, and the **Code Agent** generates a basic implementation or starter code related to the proposed methodology whenever applicable. The system utilizes RAG by extracting text from uploaded documents, converting them into embeddings, storing them in a vector database, and retrieving relevant information to generate accurate, context-aware responses.

The proposed solution significantly reduces the manual effort involved in understanding research papers while improving productivity and knowledge accessibility. It serves as an intelligent research companion for students, faculty members, and researchers by providing multiple AI-generated outputs from a single uploaded document. The project demonstrates the practical integration of **LLMs, RAG, vector databases, and multi-agent AI systems** to build an advanced document intelligence platform capable of supporting academic research and technical learning.

Keywords :

Large Language Models (LLMs) , Retrieval-Augmented Generation (RAG), Multi-Agent Systems , Artificial Intelligence , Research Paper Analysis, Document Intelligence , Vector Database , Natural Language Processing(NLP) , LangChain, CrewAI .

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References (Category specific)

1. **Lewis, P., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.**
 - <https://arxiv.org/abs/2005.11401>
2. **Yao, S., et al. (2022). ReAct: Synergizing Reasoning and Acting in Language Models.**
 - <https://arxiv.org/abs/2210.03629>
3. **Schick, T., et al. (2023). Toolformer: Language Models Can Teach Themselves to Use Tools.**
 - <https://arxiv.org/abs/2302.04761>
4. **MAIN-RAG: Multi-Agent Filtering Retrieval-Augmented Generation (ACL 2025).**
 - <https://aclanthology.org/2025.acl-long.131/>
5. **Exploration of LLM Multi-Agent Application Implementation Based on LangGraph + CrewAI (2024).**
 - <https://arxiv.org/abs/2411.18241>
6. **A Collaborative Multi-Agent Approach to Retrieval-Augmented Generation Across Diverse Data (2024).**
 - <https://arxiv.org/abs/2412.05838>
7. **MA-RAG: Multi-Agent Retrieval-Augmented Generation via Collaborative Chain-of-Thought Reasoning (2025).**
 - <https://arxiv.org/abs/2505.20096>
8. **LangChain Documentation**
 - <https://python.langchain.com/>
9. **CrewAI Documentation**
 - <https://docs.crewai.com/>
10. **LangGraph Documentation**
 - <https://langchain-ai.github.io/langgraph/>

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
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Project Guide

Name: Dr.K.Gnanendra

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INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. **All of you must chose 'Research oriented' projects only.** Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the 'Project category' section. Also provide the link to the problem statement in the 'References' section.

To write Abstract

(Use the following as a mental checklist)

- Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
- Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
- Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
- Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
- Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
- Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
- Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.